

September Exam session Projects List

Objectives

The goal of the project is to tackle one of the proposed topics in the field of Computer Vision, developing non-trivial solutions. Students should explore the problem from an original perspective, applying methods that go beyond conventional solutions and demonstrating critical thinking and problem-solving skills.

Deadline

Students willing to deliver their project in the September exam session must fill [this Google Form](#) by September 6th.

Lines of conduct

- **Student groups:** The project can be carried out by a group, each consisting of a maximum of 3 people. Projects can also be completed by individual students, but we suggest to work in team.
- **Notebook format:** The project must be implemented using a notebook (e.g., Google Colab, Kaggle) or with an IDE (e.g., VSCode, PyCharm). The code should be optimized to support GPU usage and run without any error. The delivered code must follow the structure outlined below:
 - *Imports:* all the needed packages (for the notebook format)
 - *Globals:* useful variables on the whole code
 - *Utils:* code support functions
 - *Data:* everything related to data management
 - *Network:* code to structure the neural network
 - *Train:* part containing the training cycle elements
 - *Evaluation:* tests needed for the trained network

You can find a sample template at this [link](#). Try to maintain as much as possible this conceptual structure.

- **Deep learning framework:** All projects **MUST** be done in Python via the Pytorch framework.
- **Project assignment:** You are required to choose a project through [this Google Form](#). In this form, you will provide information about your team and the project, and include the link to the project's GitHub repository. In this repository you have to upload:
 - Code (or notebook) implementing the project
 - Dataset (or a link to it)
 - Project presentation
 - Detailed README to provide a quick overview of the project and instructions on how to run it
- **Project submission:** The project must be presented on one of the exam dates. It can be presented at a different time than the written exam. Both the written exam and the project **MUST** be completed within the academic year (i.e., between the June 2026 session and the March 2027 session).
- **Plagiarism:** Any attempt to plagiarize, whether by copying other students' work, directly replicating code from online resources, or submitting content highly retrieved from generative AI models, will be strictly penalized. This course values originality and personal effort; therefore, students must submit independently developed solutions. On the other hand, it is acceptable to consult external resources for inspiration or guidance. Violations will be subject to academic disciplinary actions, potentially including a failing grade for the project or further measures outlined by university policy.

Project 1: Efficient Attention Impact on Transformer Backbones for Perceptual Similarity

Abstract: Recent research on perceptual similarity metrics has introduced DreamSim [1], a method that leverages Vision Transformer (ViT) backbones to learn embeddings aligned with human judgments of image similarity. In parallel, efficient attention mechanisms have been proposed to reduce the quadratic complexity of standard self-attention, with promising results in different vision tasks, although showing some counter-effects on maintaining the informativeness of the embeddings when applied to an encoder module [2]. This project aims to investigate whether integrating these efficient attention modules into ViT backbones within the DreamSim pipeline leads to similar effects of information dispersion and performance trade-offs.

Dataset: NIGHTS Triplets Dataset

Task: The aim of this project is to integrate efficient attention modules into ViT backbones within a DreamSim-like pipeline. The encoder will be used to compute image embeddings, and cosine distance will be used to decide which image in each triplet is more similar to a reference. By replacing standard attention layers with Meta [3], Pyra [4], or MoH [5] variants, students will assess how these modifications affect both the perceptual similarity performance (agreement with human judgments) and computational efficiency.

Main objectives:

- *Dataset preparation:* Prepare a manageable subset of the NIGHTS dataset (or similar triplets) for training and evaluation. Ensure images are preprocessed and resized to match model input requirements.
- *Model implementation:* Implement one or more ViT backbones. Integrate them into a DreamSim-like pipeline to extract embeddings and compute similarity decisions.
- *Efficient attention integration:* Replace standard self-attention layers in the backbone with efficient variants.
- *Evaluation:* Measure the agreement with human judgments (2AFC accuracy) and record inference speed on a GPU-limited environment (e.g., Colab). Compare results with the baseline backbone.
- *Ablation study:* Perform a comparative analysis across different efficient attention mechanisms and across encoder configurations (e.g., modifying all layers vs. only some layers). Discuss observed trade-offs between similarity accuracy and speed.

References:

1. Fu, S. et al. "DreamSim: Learning New Dimensions of Human Visual Similarity using Synthetic Data." NeurIPS 2023.
2. Schiavella, C. et al. "Efficient Attention Vision Transformers for Monocular Depth Estimation on Resource-Limited Hardware." Scientific Reports 2025.
3. Yu, W., Luo, M., Zhou, P., Si, C., Zhou, Y., Wang, X., . . . Yan, S. (2022). MetaFormer Is Actually What You Need for Vision. arXiv [Cs.CV]. Retrieved from <http://arxiv.org/abs/2111.11418>
4. W. Wang et al., "Pyramid Vision Transformer: A Versatile Backbone for Dense Prediction without Convolutions," 2021 IEEE/CVF International Conference on Computer Vision (ICCV), 2021.
5. Jin, P., Zhu, B., Yuan, L., & Yan, S. (2025). MoH: Multi-Head Attention as Mixture-of-Head Attention. arXiv [Cs.CV]. Retrieved from <http://arxiv.org/abs/2410.11842>

Project 2: Out of Distribution Convolutional Response for Monocular Depth Estimation

Abstract: Recent advances in out-of-distribution (OOD) detection have introduced novel scoring methods based on the analysis of convolutional responses in deep neural networks. In particular, the CORES (Convolutional Response-based Score) method [1] exploits the magnitude and frequency of extreme activations in convolutional kernels to discriminate between in-distribution and OOD inputs without requiring access to the original training data. This project aims to explore the applicability of the CORES methodology within the context of Monocular Depth Estimation (MDE), a dense prediction task typically addressed through fully convolutional encoder-decoder architectures. This project aims to enable a clearer and more interpretable analysis of convolutional responses and their potential as OOD indicators.

Dataset: NYU Depth v2 (indoor scenes), KITTI (outdoor scenes)

Task: The aim of this project is to integrate the CORES scoring mechanism into a convolutional-base MDE network. The chosen network will be trained on an in-distribution dataset (e.g., NYU Depth v2) and subsequently evaluated on both in-distribution and out-of-distribution samples. The study will assess whether convolutional response patterns can serve as reliable OOD indicators in a dense prediction setting, using metrics such as AUROC and FPR95.

Main objectives:

- *Dataset preparation:* Preprocess and split the NYU Depth v2 dataset for training and validation. Collect OOD samples (e.g., KITTI images resized to match the input resolution). As an alternative, consider using a pretrained generative model to synthesize OOD samples.
- *Model implementation:* Implement and train a lightweight depth estimation model. Some recommended architectures are listed in references [2,3], but other models can also be considered. Training and comparing two or more models is encouraged.
- *CORES integration:* Integrate the convolutional response scoring method to extract layer-wise response statistics and compute the OOD score.
- *Evaluation:* Assess OOD detection performance (AUROC, FPR95) and depth estimation accuracy (RMSE, Abs_{Rel} , δ_1 , δ_2 , δ_3) on both ID and OOD datasets.
- *Ablation study:* Perform a comparative analysis across different layers and convolutional architectures, and evaluate the impact of model depth on the observed response patterns.

References:

1. Tang, Keke, et al. "CORES: Convolutional Response-based Score for Out-of-distribution Detection." CVPR 2024.
2. Wofk, Diana, et al. "FastDepth: Fast Monocular Depth Estimation on Embedded Systems." ICRA 2019.
3. Papa, L., Russo, P., & Amerini, I. "METER: A Mobile Vision Transformer Architecture for Monocular Depth Estimation". IEEE Transactions on Circuits and Systems for Video Technology, 2023.

Project 3: Anatomically-Constrained LDMs for Rare Case Augmentation

Abstract: Deep learning models for Periodontitis often suffer from severe class imbalance, as "Stage IV" (severe) cases are significantly rarer than "Stage I" cases in public datasets. While Generative Models can synthesize data, standard Denoising Diffusion Probabilistic Models (DDPMs) often suffer from anatomical hallucinations, where the generated images look realistic but lack clinically correct tooth and bone structures. This project investigates how to use Latent Diffusion Models to generate rare clinical cases that are strictly constrained by anatomical masks to ensure clinical validity.

Dataset: **Periodontal Bone Loss Keypoint and Detection Dataset** (includes Bounding Boxes, Keypoints, and Instance Segmentation masks).

Task: The project involves developing a 2D-LDM pipeline to synthesize high-quality dental X-ray patches for severely underrepresented classes. Students will implement methodological enhancements to ensure the generated textures align with real-world anatomical structures. The work includes designing a guidance mechanism, such as a mask-conditioning branch using segmentation masks or an anatomical-based loss, to keep bone levels and lesions geometrically consistent with the tooth root. The effectiveness of the synthetic data will be validated through standard generative models evaluation metrics and by measuring its impact on a downstream classification and/or detection model.

Main objectives:

- *Baseline LDM Implementation:* Implement a standard 2D Diffusion model (e.g. LDM) trained on localized, high-resolution cropped patches of rare lesions, to generate and identify common structural errors in unconstrained generation.
- *Anatomical Guidance Integration:* Formulate and integrate an original guidance mechanism to maintain strict clinical consistency between the generated bone levels and actual anatomical structures.
- *Downstream Impact Evaluation:* Artificially augment the training set by importing the synthesized, anatomically-correct minority class in the imbalanced dataset. Compare the Mean Average Precision (mAP) of a downstream model trained on real data against the (real+)augmented data, evaluating on a strict holdout test set.
- *Ablation Studies:* Conduct an ablation study to analyze the trade-off between the strength of the anatomical mask constraints and the diversity of the generated imagery, as well as a possible comparison with a DDPM.

References:

- 1 Ho, J., Jain, A., & Abbeel, P. "Denoising Diffusion Probabilistic Models". NeurIPS 2020.
- 2 Kazerooni, A., et al. "Diffusion models in medical imaging: A comprehensive survey", Medical Image Analysis 2023. DOI: 10.1016/j.media.2023.102846
- 3 Pinaya, W. H. L., et al. "Brain Imaging Generation with Latent Diffusion Models". MICCAI 2022.
- 4 Konz, N., et al. "Anatomically-Controllable Medical Image Generation with Segmentation-Guided Diffusion Models". MICCAI 2024.

Project 4: Emerging Interpretability in LeJEPA vs. Supervised Learning

Abstract: This project investigates the emergence of "zero-shot" semantic segmentation in Joint-Embedding Predictive Architectures (JEPAs) compared to traditional supervised models. While supervised training is driven by categorical labels, LeJEPA utilizes Sketched Isotropic Gaussian Regularization (SIGReg) to learn representations. We aim to evaluate whether the semantic structures emerging in LeJEPA's latent space (visible via PCA) align more naturally with the model's internal attention and gradients than those in supervised counterparts.

Dataset: ImageNet-1K, ImageNet-100, CIFAR-10, CIFAR-100.

Task: A comparative layer-by-layer analysis of feature localization and semantic consistency between LeJEPA-trained models and End-to-End Supervised models, utilizing both Vision Transformers (ViT) and Convolutional Neural Networks (CNNs).

Main objectives:

- *Model Training and Feature Extraction:* Deploy and contrast ViT and CNN architectures trained via LeJEPA and standard supervised learning. Extract latent representations across different layers for comparative probing.
- *Mapping XAI and Semantic Projections:* Implement a retrieval pipeline for explainable ai (XAI) saliency maps (using GMAR or Attention Rollout for ViTs and Grad-CAM for CNNs) and compute semantic masks through PCA projections of the embedding space.
- *Definition of a Semantic Alignment Score (SAS):* Design and implement a quantitative framework to measure the association between XAI-based saliency and PCA-based semantic components. The student should explore and justify the choice of metrics (e.g., spatial correlation, mutual information, or localized overlap) to define a robust score.
- *Qualitative and Comparative Analysis:* Conduct an in-depth visual study to assess if LeJEPA's "free" semantic segmentation provides a more interpretable representation of objects compared to the potentially biased focus of supervised models.

References:

- Balestrieri, R., & LeCun, Y. (2025). *LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics*.
- Jo, S., Jang, G., & Park, H. (2025). *GMAR: Gradient-Driven Multi-Head Attention Rollout for Vision Transformer Interpretability*.
- Selvaraju, R. R., et al. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*.

Project 5: Compressed-Domain Deepfake Attribution via JPEG AI Latents

Abstract: Modern image analysis pipelines for deepfake forensics universally begin by fully decoding compressed images into pixel arrays. JPEG AI, the recently standardised learned image compression codec, encodes images into structured latent tensors using deep neural networks. Unlike the hand-crafted DCT coefficients of classical JPEG, these latents have been shown to carry rich semantic information and, crucially, generator-specific statistical signatures that survive compression. This project asks students to investigate and extend this paradigm: can compressed-domain representations, operating before full decoding, serve as effective and scalable descriptors for deepfake attribution? Students are expected to assess the representational capacity of JPEG AI latents, explore lightweight attribution pipelines that operate in this compressed space, and propose an original contribution that pushes performance, robustness, or scope beyond the proof-of-concept established in the literature.

Dataset: **WILD (Wild Image Linkage Dataset)**, which provides synthetic images from multiple state-of-the-art generative models (diffusion models and GANs) along with corresponding provenance metadata.

Task: The project involves building a deepfake attribution pipeline that operates primarily in the compressed domain of JPEG AI, rather than on decoded pixel data. Students will compress the dataset images using the JPEG AI reference encoder, extract the resulting latent tensors, and use these representations to perform generator source attribution. The attribution task requires assigning a synthetic image to its specific generative model (e.g., distinguishing StyleGAN2 from Stable Diffusion from Adobe Firefly). Students are expected to establish a baseline pipeline, critically evaluate its limitations, and formulate an original improvement addressing at least one identified weakness.

Main objectives:

- *Compressed-Domain Baseline:* Implement an attribution pipeline that extracts descriptors from JPEG AI latent tensors and performs source attribution. Reproduce and document baseline results across multiple generators, reporting standard retrieval and attribution metrics.
- *Representational Analysis:* Investigate the structure of the JPEG AI latent space as it pertains to forensics. Assess how attribution performance degrades as the number of generators increases and how JPEG AI compression bits-per-pixel affect the performance.
- *Robustness Evaluation:* Evaluate the attribution pipeline under realistic post-processing conditions (e.g., JPEG recompression, resizing, social media-style transformations). Quantify how the forensic fingerprints embedded in the latent space are affected by such operations.
- *Original Contribution:* Propose and implement a methodological improvement to address a chosen limitation of the baseline. The specific direction is left to the students' discretion: improvements to the descriptor extraction strategy, the attribution mechanism, the handling of unseen generators, or the efficiency of the pipeline are all valid avenues.

References:

- 1 Ragaglia, C.V., Catania, L., Guarnera, F., Allegra, D., & Battiato, S. (2025). *What if Retrieval Could Work Before Decoding? The case of JPEG AI Latents for Deepfake Source Attribution*. DFF '25, ACM. <https://doi.org/10.1145/3746265.3759672>
- 2 Ascenso, J., Alshina, E., & Ebrahimi, T. (2023). *The JPEG AI standard: Providing efficient human and machine visual data consumption*. IEEE Multimedia, 30(1), 100–111.
- 3 Cannas, E.D., et al. (2025). *Is JPEG AI going to change image forensics?* arXiv:2412.03261.
- 4 Bongini, P., et al. (2025). *WILD: A new in-the-Wild Image Linkage Dataset for synthetic image attribution*. arXiv:2504.19595.

- 5 Guarnera, L., Giudice, O., & Battiato, S. (2024). *Mastering deepfake detection: A cutting-edge approach to distinguish GAN and diffusion-model images*. ACM TOMM, 20(11).

Project 6: Frequency-Domain Diffusion Models: Synthesizing Spectral Fingerprints for Deepfake Analysis

Abstract: Deepfakes and AI-generated images, despite achieving impressive photorealism in the spatial domain, consistently exhibit structural artifacts and tell-tale "fingerprints" in the frequency domain. These spectral anomalies are widely exploited by state-of-the-art deepfake detectors. This project explores a novel generative paradigm: training Diffusion Models directly on the frequency spectrum (e.g., via Discrete Cosine Transform or Fourier Transform) rather than the standard RGB spatial domain. By modeling the true spectral distribution of real images, the goal is to synthesize realistic spectra that, once inversely transformed, produce coherent RGB images. Students will investigate whether this approach can bridge the gap between generation and forensics by synthesizing images devoid of traditional generative frequency artifacts.

Dataset: FaceForensics++, Celeb-DF, GenImage, or standard high-quality facial datasets like FFHQ/CelebA. (*Or what you can run on your hardware*)

Task: The objective is to implement a pipeline that transforms a dataset of images into their frequency representations and trains a baseline Diffusion Model (DDPM) to generate these spectral maps from noise. Students must implement the forward and inverse frequency transforms, adapt the UNet architecture to handle complex or multi-channel frequency data (e.g., magnitude/phase or real/imaginary components), and reconstruct spatial RGB images from the generated spectra. Otherwise implement your own pipeline coherently with the idea of the proposed project. It is required to evaluate both the visual quality of the reconstructed images and their spectral properties compared to standard RGB-trained models.

Main objectives:

- *Frequency Transformation Pipeline:* Implement a robust preprocessing step to convert spatial images to the frequency domain (e.g., 2D-DCT or 2D-FFT) and format the data appropriately for neural network processing.
- *Anatomical Guidance Integration:* Formulate and integrate an algorithmic improvement (e.g., DDIM sampling strategy, a latent-space approach, progressive distillation, or something new!) aimed at accelerating inference or training.
- *Spectral Diffusion Training:* Train a baseline diffusion model entirely in the frequency domain. Address architectural challenges, such as balancing the high-variance low-frequency components with the sparse high-frequency details.
- *Reconstruction & Visual Fidelity:* Invert the generated spectra back to the spatial domain. Evaluate generation quality using standard quantitative metrics (FID, IS) and visually inspect the structural coherence of the resulting RGB images.
- *Forensic Spectral Analysis:* Compare the frequency spectra of the images generated by your model against those generated by a standard spatial DDPM. Conduct a comparative analysis to determine if the frequency-domain model successfully mitigates the typical grid-like artifacts or "synthetic fingerprints" common in spatial generative models.

Otherwise implement your own pipeline coherently with the idea of the proposed project.

References:

- 1 Frank, J., et al. "Leveraging Frequency Analysis for Deep Fake Image Recognition." ICML 2020.
- 2 Corvi, R., et al. "On the detection of synthetic images generated by diffusion models." ICASSP 2023.

- 3 Xu, L., et al. "Text-to-image with frequency domain diffusion models." SPIE 2025.
- 4 Song, T., et al. "FilterDiff: Noise-free Frequency-domain Diffusion Models for Accelerated MRI Reconstruction." MICCAI 2025.

Project 7: TransoNet, Lightweight Transformer Occupancy Networks for 3D virtual object reconstruction

Abstract: In the context of augmented, virtual reality and computer graphics, the 3D object reconstruction task from a sparse point cloud is highly computationally demanding and for this reason, it is difficult to accomplish on embedded devices. In order to find the best balance between time for mesh generation and mesh quality, we aim to tackle the object reconstruction process by developing a lightweight implicit representation. To achieve this goal, we leverage the use of Convolutional Occupancy Networks.

Dataset: ShapeNet.

Task: Enhance the model's performance by analyzing its architecture and applying lightweight optimization methods. The model consists of three main components: an Encoder, a Dynamic ViT, and a Decoder. Students will evaluate some or all of these components to identify architectural bottlenecks and subsequently implement modifications to improve overall network efficiency.

Main objectives:

- *Detect the bottleneck:* Conduct an architectural analysis using methods such as pruning, MAC (Multiply-Accumulate) counting, memory usage, etc. The goal is to pinpoint the bottleneck and empirically demonstrate its location.
- *Application of lightweight methods:* Once the bottleneck is identified, utilize lightweight techniques or directly modify the selected component to enhance its performance.
- *Ablation study:* Compare the model's performance before and after the proposed modifications.
- *Efficiency-Accuracy Trade-off:* Quantify the performance gains relative to the resulting mesh quality, using the metrics detailed in the repository's linked paper.

References:

- Lionar, S., Emtsev, D., Svilarkovic, D. & Peng, S. (2020). *Dynamic Plane Convolutional Occupancy Networks*.
- Wang, Y., Huang, R., Song, S., Huang, Z., & Huang, G. (2021). *Dynamic Transformers for Efficient Image Recognition*.
- Melis Tonti, C., & Amerini, I. (2025). *Lightweight Transformer Occupancy Networks for 3D virtual object reconstruction*.
- Melis Tonti, C., Papa, L., & Amerini, I. (2024). *Lightweight 3-D Convolutional Occupancy Networks for Virtual Object Reconstruction*.

Project 8: Self-Supervised Latent Representations for Imbalanced Apical Periodontitis Grading

Abstract: Automated grading of Apical Periodontitis in panoramic radiographs is a challenging clinical task often plagued by class imbalance, as severe lesions (PAI 5) are significantly rarer than mild structural changes (PAI 3) in public data pools. While supervised models overfit to the majority class, self-supervised Joint-Embedding Predictive Architectures (JEPAs) offer a robust alternative. By predicting the latent representations of masked anatomical patches rather than reconstructing pixels, JEPAs learn rich, invariant structural semantics from the entire dataset without human labels. This project investigates whether a Vision-JEPA model pre-trained on unannotated dental X-rays can cultivate an explicit understanding of bone density and lesions, thereby allowing a lightweight downstream classifier to generalize to rare, severe pathology blocks even when trained on a highly skewed distribution.

Dataset: **Panoramic Radiographs with Periapical Lesions Dataset** A dataset of 3,926 original panoramic images containing 6,029 labeled lesions. Students must handle validation/testing splits strictly at the patient level to avoid data leakage.

Task: The project involves implementing a self-supervised Vision-JEPA (e.g., I-JEPA or LeJEPA) using a Vision Transformer (ViT) backbone. In the pre-training phase, the network will process unannotated, resized panoramic images, learning to predict the embeddings of missing patches from localized contexts. In the downstream phase, the pre-trained encoder weights will be frozen. Using the provided bounding box coordinates from the dataset's XML files, students will extract the latent vectors corresponding to the lesion areas and train a lightweight multi-class classification head to predict the PAI grade (3, 4, or 5). Students will formulate an original methodological novelty (e.g., a cost-sensitive latent loss, focal attention blocks, or balanced token-sampling strategies) to handle the severe scarcity of the PAI 5 class.

Main objectives:

- *Self-Supervised Architecture Design:* Implement a complete PyTorch-based Vision-JEPA pipeline consisting of a Context Encoder, a Target Encoder updated via Exponential Moving Average (EMA), and a shallow Predictor network.
- *Imbalance-Aware Downstream Classification:* Integrate a lightweight attention pooling layer and a classification head over the frozen JEPA features. Train the downstream classifier on the heavily skewed distribution (3,691 PAI 3 vs. 521 PAI 5 cases).
- *Methodological Novelty Integration:* Formulate and integrate an original algorithmic novelty specifically designed to combat class imbalance in the latent space (e.g., asymmetric embedding regularizers or class-balanced patch masking).
- *Ablation Studies:* Conduct an ablation study evaluating the impact of the proposed novelty. Performance on the minority class should be reported using threshold-agnostic metrics (e.g., Macro-F1, Precision-Recall AUC, and confusion matrices) rather than global accuracy.

References:

- Assran, M., et al. (2023). *Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture*. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).
- Balestrieri, R., & LeCun, Y. (2025). *LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics*. arXiv preprint.
- Do, H. V., et al. (2024). *A Dataset of apical periodontitis lesions in panoramic radiographs for deep-learning-based classification and detection*. Data in Brief, 54, 110486.

Project 9: Content-Sensitivity-Aware Deepfake Detection

Abstract: The rapid advancement of generative AI models has dramatically narrowed the visual gap between synthetic and authentic imagery, rendering traditional deepfake detectors increasingly unreliable. A critical yet underexplored dimension of this problem is the *sensitivity* of the depicted content: a manipulated image of an everyday scene carries vastly different societal implications from a fabricated image of a political figure, a war zone, or a public health emergency. This project addresses this gap by requiring students to build a novel benchmark dataset pairing real and synthetic images across multiple levels of content sensitivity, and to leverage this sensitivity signal to develop a more context-aware approach to deepfake detection.

Dataset: Students must construct their own dataset by curating and combining images from publicly available sources. The resulting dataset must span a meaningful spectrum of content sensitivity levels, from low-sensitivity everyday imagery to high-sensitivity scenarios such as political events, conflict zones, and public health crises. Both real and AI-generated images must be included at each sensitivity level.

Task: The project is organized around two interconnected goals. The first is the curation of a sensitivity-stratified benchmark for deepfake detection, where each image is annotated with both an authenticity label (real/fake) and a content sensitivity score. The second is the development of a detection approach that explicitly incorporates content sensitivity into its decision process, with the aim of improving reliability on the most critical and high-stakes content. Students are free to explore the design space of the detection model, which may draw from discriminative classifiers, vision-language reasoning, or hybrid architectures.

Main objectives:

- *Dataset Construction:* Collect and organize images from multiple existing datasets, assigning each image a content sensitivity label. Students must define a clear and justified sensitivity taxonomy and document their curation methodology. The assignment of sensitivity scores should be systematic and reproducible.
- *Sensitivity-Stratified Benchmarking:* Evaluate one or more pre-trained deepfake detectors on the assembled dataset. Analyse how detection performance varies across sensitivity strata, and introduce an evaluation metric that accounts for the sensitivity of the content being examined.
- *Sensitivity-Aware Detection:* Design and implement a detection approach that integrates content sensitivity into the deepfake classification pipeline. The specific technical solution is left to the students' creativity: the key requirement is that the model's predictions are meaningfully informed by the sensitivity of the input content.
- *Ablation and Discussion:* Quantify the contribution of the sensitivity component through ablation experiments. Discuss the ethical implications of sensitivity-aware moderation systems and the limitations of the proposed approach.

References:

- 1 Tan, S., et al. (2025). *RRDataset: A Large-Scale Benchmark for Robust Real-World Deepfake Detection*. arXiv:2509.09172.
- 2 Shao, R., et al. (2023). *DGM4: Detecting and Grounding Multi-Modal Media Manipulation*. arXiv:2304.02556.
- 3 Xia, Z., et al. (2024). *SID-Set: A Social-Media-Oriented Dataset for Deepfake Detection*. arXiv:2412.04292.
- 4 *OpenFAKE: A Large Politically Grounded Deepfake Benchmark*. arXiv:2509.09495.